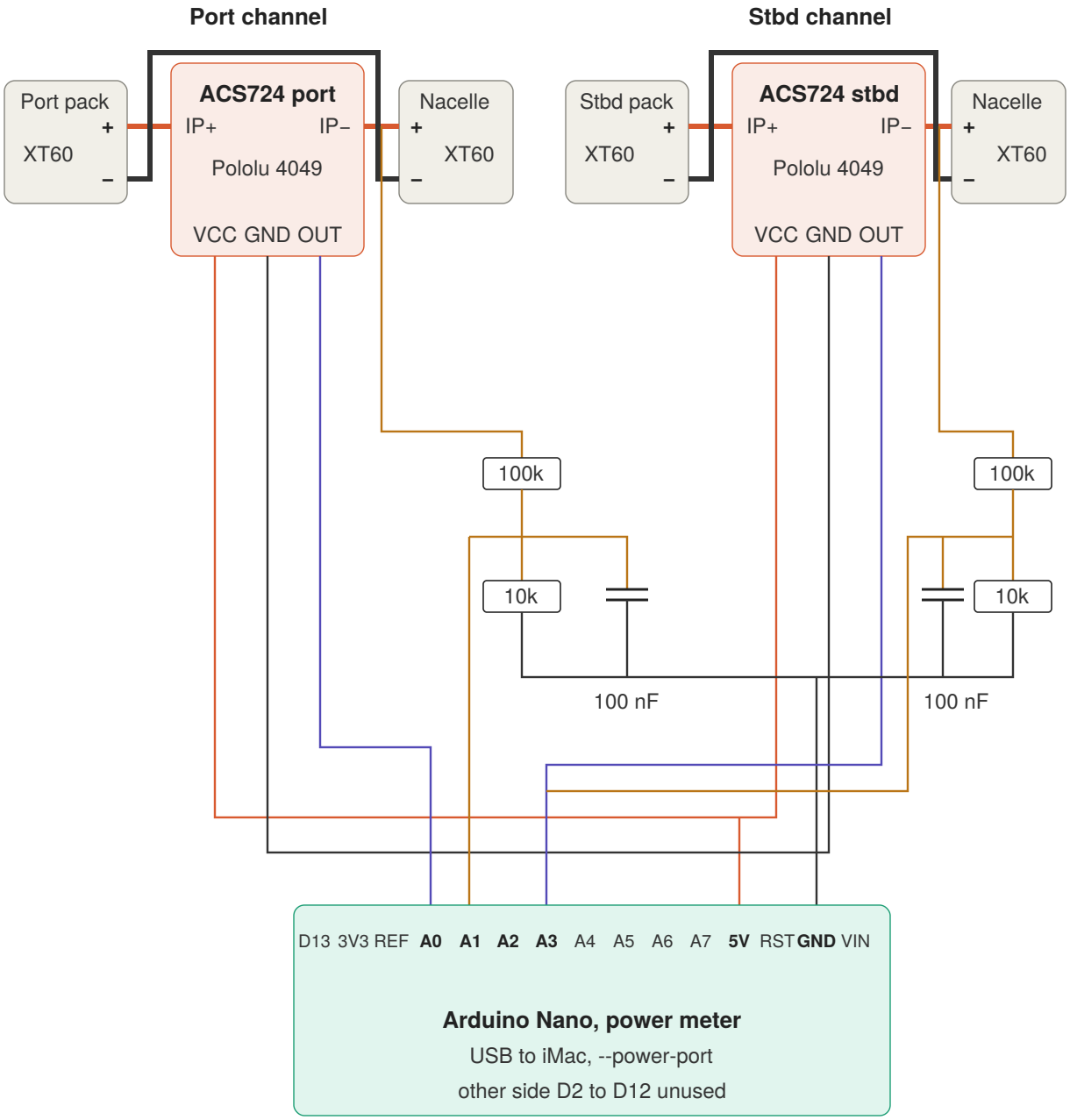
Power Nano, two-channel ACS724 schematic



Legend

- Pack positive, 12 AWG, through IP+ then IP–, board arrow toward nacelle
- Pack negative, 12 AWG, straight pack to nacelle, never on the sensor board
- VCC, red 22 to 24 AWG, both boards to the one Nano 5V pin
- GND, black 22 to 24 AWG, both boards and both dividers to Nano GND
- OUT, current signal: port to A0, stbd to A2
- Voltage sense: 100k from nacelle-side positive, 10k to GND, junction to A1 or A3

100 nF ceramic cap from each junction to GND, on the perfboard at the Nano. FILTER holes stay empty.

Nano top row as printed: D13, 3V3, REF, A0 ... A7, 5V, RST, GND, VIN. Nothing on the other row.